Chenglong Hua

Henan, China
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Skills and tools

CAD: SolidWorks, AutoCAD, Design for assembly, Design for manufacturing, UAV design
Simulation: ANSYS Fluent, COMSOL
Material characterization: SEM, XRD, super depth-of-field microscopy
Data analysis: Origin, Visio

Education Background

BEng (Hons) Aerospace Engineering (Year 4) 09/2022-Present
University of Nottingham Ningbo China (UNNC) Ningbo, China
GPA: 4.0/4.0; Ranked 2nd in cohort both year; Outstanding Student Award (Top 2%)
Selected core modules: Aerospace Electr Eng & Compu (81), Aircraft Design and Performance (81), Aerospace Design and Manufacture (81), AERO EE & Applications (80), et al.

Project Experience

Diamond Dresser Design via Laser-Cut Pyramid Arrays 12/2024-Present
Group Leader of the Research Project, Advanced Manufacturing Centre, UNNC Ningbo, China

- **Experimental design**: conducted surface micromorphology characterisation of polishing pad samples using SEM under controlled tool pressure and polishing fluid flow conditions; compared results with samples produced by commercial CVD equipment.
- **Material characterisation**: analysed polishing pad surface morphology through SEM imaging to evaluate surface roughness and identify porosity features.

Blade-Structured Grinding Tool Based Air-Cooling Strategy 06/2025-Present
Advanced Manufacturing Centre, UNNC Ningbo, China

- **Mechanical design**: designed and optimised the blade structures of a surgical grinding tool.
- **Experimental design**: conducted surface characterisation of tooth samples after one hour of grinding at 20,000 rpm, comparing the new design with conventional water-cooled grinders.
- **Characterisation**: assisted in thermal testing using thermocouple arrays to measure tool temperature during grinding; supported post-grinding surface analysis of tooth samples using SEM and super depth-of-field microscopy.

ML-Enhanced Dissolution Forecasting for Solid Dosage Forms 07/2025- Present
Advanced Manufacturing Centre, UNNC Ningbo, China

- **Modelling:** established the physical model of the pills and processing and visualizing the data.

National UAV Design Competition-Micro Fixed-Wing Aircraft 02/2025-06/2025
Spring Research Project, UNNC Ningbo, China

- **Mechanical design**: designed micro aircraft within strict dimensional constraints using S5010 airfoil and large flying-wing structure.
- **Simulation modelling**: modelled in XFLR5 and Inventor; advanced to national finals.

Laser Thinning of Silicon Wafers via Diffractive Optical Elements 08/2024-02/2025
Y-Lab, UNNC Ningbo, China

- **Literature review**: completed a survey with a comprehensive technical report over 80 papers covering six major wafer thinning processes, surface roughness and material subtraction.
- **Product design**: supported early-stage process design for rotational laser-based wafer thinning.

Flexible Pressure Sensor for Automated Acne Scar Treatment 06/2024-09/2024
Advanced Manufacturing Centre, UNNC Ningbo, China

- **Electronic design and testing:** developed soft pressure sensors and conducted 50+ pressure-electrical signal calibration tests.

- **Mechanical & Mechatronic design:** Designed an integrated system for automated drug delivery targeting scarred skin areas.

Laser-Induced Graphene Fabrication on Nomex Paper 06/2024-08/2024
Summer Research Project, UNNC Ningbo, China

- **Troubleshooting:** detected and raised the problem that the lab-made paper graphene sensor shows greater resistance when bent than when flat
- **Experiment Design**: to evaluate the response accuracy, the experiment measured the electrical resistance over time using a commercial infra pyrometer; to evaluate the duration of the graphene paper, fatigue tests of 1000 circle has been performed.
- **Data Analysis**: identified optimal process window of the fabrication process via analysing fatigue and electrical resistance measurement data.

Publication

In Preparation:
Li, Q., **Hua, C.,** Wang, B., Zheng, Z., Li, K., Xu, Y., Liu, X., Zhao, Y., Liu, G., Li, H., A Novel Single-Crystal Diamond Dresser: Fabrication and Performance Evaluation, *Journal of Manufacturing Processes*, 2025.
Submitted or Under Reviewed:
Ma, Y., Tong, Y., Jiang, X., **Hua, C.,** Zhai, Y., Wildman, R., Li, H., He, Y., A Feature-Based Machine Learning Approach for Rapid Prediction of Drug Release from 3D-Printed Pill Geometries, *International Journal of Pharmaceutics*, 2025.
Published:
Li, Q., Wang, Y., Xiu, H., Li, K., Wang, B., **Hua, C**., Zhao, Y., Liu, G. and Li, H.N., 2025. Rotary laser thinning of silicon wafer via diffractive optical element: Effects of process parameters on surface integrity. Journal of Manufacturing Processes, 143, pp.160-177.
Li, K., Liu, G., Li, Q., Tong, C., Chai, W.S., **Hua, C**., Wei, W., Jiang, J., Zhao, Y., Low, S.S. and Li, H.N., 2025. The fabrication of laser-induced graphene on nomex paper: Effects of process parameters and flexible wearable applications. Optics & Laser Technology, 190, p.113228.

Patent

Co-developed, "Paper-Based Laser-Induced Graphene Sensor Apparatus" - (Pending) CN Patent Application No.: CN202422639952.X

Internship Experience

Engineering Intern 01/2024-02/2024
Zhiwei Technology Co., Ltd. Ningbo, China

- Participated in process improvement for polyurethane foam production.
- Performed material characterization including density, thickness, and hardness measurements to support quality optimization.

Extracurricular Activities

President Since 10/2024
Aerospace Society, University of Nottingham Ningbo China
Committee Member Since 04/2024
RAeS China Youth Branch
Member
Royal Aeronautical Society Since 03/2024

Awards

Language Skills: English (Professional), Mandarin (Native)
Honors & Awards:
National Scholarship, 2025
UNNC Outstanding Student, 2025
UNNC Outstanding Student Leader, 2025
Provost's Scholarship, 2024-2025
Volunteer, International Symposium on Case Studies in Advanced Engineering Design 2024